

Racial and Gender Inequality Discourses in Academic Research: Dataset Documentation

1 Overview

We constructed the empirical corpus from a Semantic Scholar database maintained in SOLR by Professor Philip M. Parker’s data team at INSEAD. The database contains approximately 180 million research papers with metadata including titles, abstracts, authors, publication venues, and publication years.

Using expert-generated keyword dictionaries, we identified papers published between 2005 and 2021 that contained at least one racial- or gender-inequality keyword in the title or abstract. We also sampled approximately 50,000 papers containing no keywords from either dictionary as a shared negative set.

This procedure yielded two classification datasets:

- `race/race_ineq_dataset.csv`: approximately 80,000 papers, comprising papers containing racial-inequality keywords and the shared negative set.
- `gender/gender_ineq_dataset.csv`: approximately 95,000 papers, comprising papers containing gender-inequality keywords and the same shared negative set.

Each row represents one academic paper. The two datasets share the same column structure. Feature names correspond to those used in the associated figures and machine-learning analyses.

2 Data Sources and Provenance

The source text retrieved from the Semantic Scholar database consists of paper titles and abstracts. All other variables described below are derived or constructed features.

3 Important Note on Demographic Proxy Variables

Several variables in these datasets are constructed using probabilistic demographic inference from author names and/or country-level population distributions. These variables are probabilistic proxies derived from names and/or country-level population distributions and should not be interpreted as direct measures of individuals’ self-identified gender, race, ethnicity, or nationality.

4 Dataset Column Dictionary

4.1 Text

Column	Type	Description	Used in ML analysis
<code>title</code>	string	Paper title.	no
<code>paper_abstract</code>	string	Paper abstract.	no

4.2 Label

Column	Type	Description	Used in ML analysis
<code>count_inequality_words</code>	int	How many inequality-related words appear in <code>title</code> + <code>paper_abstract</code> (whole-word matches). Classification label is positive when this count is greater than 0.	yes (label)

4.3 Author Gender Proxies

Name-based estimates of how likely each author is female (probability on a 0–1 scale), then aggregated across the author list.

Column	Type	Description	Used in ML analysis
<code>num_authors</code>	int	Number of authors on the paper.	yes
<code>female</code>	float (0–1)	Mean female probability across all authors.	yes
<code>female_max</code>	float (0–1)	Maximum female probability among coauthors.	no
<code>female_min</code>	float (0–1)	Minimum female probability among coauthors.	no
<code>first_author_female_score</code>	float (0–1)	Female probability of the first author only.	no

4.4 Fields of Study

Whether the paper falls into each coarse disciplinary group. A paper may have multiple 1s.

Column	Type	Description	Used in ML analysis
natural_science	0/1	1 if the paper is in natural sciences (e.g., biology, chemistry, physics, geology, geography, environmental science, medicine, materials).	yes
engineering_and_technology	0/1	1 if the paper is in engineering / technology / mathematics (e.g., engineering, computer science, mathematics).	yes
social_science	0/1	1 if the paper is in social sciences or humanities (e.g., sociology, political science, psychology, economics, history, art, philosophy).	yes

4.5 Country-Level Racial Diversity

For each author, a likely country of origin is inferred from the name; that country's racial population shares are used to compute diversity metrics.

Column	Type	Description	ML
country_race_diversity_score	float	Mean Shannon entropy of racial shares in authors' countries, $H = -\sum_i p_i \log(p_i)$. Higher values indicate greater diversity and are sensitive to both rare groups and evenness.	yes
country_race_simpson_index_mean	float	Mean Simpson diversity of racial shares in authors' countries, $1 - \sum_i p_i^2$. Higher values indicate greater diversity.	no
country_race_inverse_dominance_mean	float	Mean inverse dominance of racial shares in authors' countries, $1/\max(p_i)$. Higher values indicate less single-group dominance.	no

4.6 Paper-Level Racial Diversity

Column	Type	Description	ML
authors_race_diversity_score	float	Shannon entropy of the paper’s four-group (Asian / Black / White / Hispanic) racial composition. Higher values indicate greater diversity.	yes
paper_race_simpson_index	float	Simpson diversity of the paper’s four-group composition, $1 - \sum_i p_i^2$. Higher values indicate greater diversity.	no
paper_race_inverse_dominance	float	$1/\max(p_i)$ for the paper-level composition. Higher values indicate less single-group dominance.	no
white	float (0–1)	Mean White population share of authors’ countries.	yes
asian	float (0–1)	Mean Asian population share of authors’ countries.	yes
black	float (0–1)	Mean Black population share of authors’ countries.	yes
hispanic	float (0–1)	Mean Hispanic population share of authors’ countries.	yes

4.7 Year-Level Inequality Discourse Context

How common inequality-related language was in the broader academic literature or news around the paper’s publication year t .

4.7.1 Inequality Language in Academic Literature

Column	Description	ML
acad_ineq_t-0	Relative frequency (mentions relative to corpus size) of inequality language in academic literature in publication year t .	no
acad_ineq_t-1	Same measure in year $t - 1$.	no
acad_ineq_t-2	Same measure in year $t - 2$.	no
acad_ineq_t-3	Same measure in year $t - 3$.	no
paper_inequality_mentions_3_years	Mean of the academic inequality-language measure over years $t - 2$, $t - 1$, and t .	yes

4.7.2 Inequality Language in News

Column	Description	ML
news_ineq_t-0	Relative frequency of inequality language in news in publication year t .	no
news_ineq_t-1	Same measure in year $t - 1$.	no
news_ineq_t-2	Same measure in year $t - 2$.	no
news_ineq_t-3	Same measure in year $t - 3$.	no
news_inequality_mentions_3_years	Mean of the news inequality-language measure over years $t - 2$, $t - 1$, and t .	yes

4.7.3 News — Gender / Economic / Race Subsets

Column	Description	ML
news_gender_ineq_t-0	Relative frequency of gender-inequality language in news in publication year t .	no
news_gender_ineq_t-1	Same measure in year $t - 1$.	no
news_gender_ineq_t-2	Same measure in year $t - 2$.	no
news_gender_ineq_t-3	Same measure in year $t - 3$.	no
news_gender_ineq_3yr_avg	Mean over publication-year window $t - 2$, $t - 1$, and t .	no
news_econ_ineq_t-0	Relative frequency of economic-inequality language in news in publication year t .	no
news_econ_ineq_t-1	Same measure in year $t - 1$.	no
news_econ_ineq_t-2	Same measure in year $t - 2$.	no
news_econ_ineq_t-3	Same measure in year $t - 3$.	no
news_econ_ineq_3yr_avg	Mean over publication-year window $t - 2$, $t - 1$, and t .	no
news_race_ineq_t-0	Relative frequency of race-inequality language in news in publication year t .	no
news_race_ineq_t-1	Same measure in year $t - 1$.	no
news_race_ineq_t-2	Same measure in year $t - 2$.	no
news_race_ineq_t-3	Same measure in year $t - 3$.	no
news_race_ineq_3yr_avg	Mean over publication-year window $t - 2$, $t - 1$, and t .	no

5 Citation

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